University
Mohammed VI
Polytechnic

College of Computing

Deliverable # 5: Title of Lab/Project here (e.g.: Conceptual Design ....)

Data Management Course

UM6P College of Computing

Professor: Karima Echihabi **Program:** Computer Engineering

Session: Fall 2026

Team Information

Team Name	UNO
Member 1	Daniel Tchonang
Member 2	Amine Sira
Member 3	Mohammed Oumane
Member 4	Zaid Zehari
Repository Link	`https://github.com/unocomputing/unodatabase.git`

1 Roles

- Daniel: Solving problems
- Sira: Report + solving problems
- Mohammed: solving problems
- Zaid; Solving problems

2 Question 1

all FDs in every relations are the key determining the table, the key is a superkey so they are all BCNF.

3 Question 2

A)

- The only candidate key is Serial No
- The relation is NOT BCNF
- The FD Model Id → manufacturer violates BCNF. Model Id is not a superkey because it only determines itself and the manufacturer. It is also not 3NF because manufacturer is not part of the candidate key. Since Model Id is atomic, we can conclude that the strongest form is 1NF
- Decomposition: R1= (serial No, model Id ,lab Id), R2= (model Id, manufacturer)
- Decomposition is lossless because model Id (the intersection between R1 and R2) is a key of R2
- FDs are preserved: serial No → model Id, lab Id holds in R1 and model Id → manufacturer holds in R2

B)

- Candidate key: (pid, projectCode). A person can participate in several projects, and a project can have several people, so both attributes are needed to identify a participation.
- BCNF: No. projectCode → piPid violates BCNF because projectCode does not identify a full row in ProjectParticipation.
- 3NF: No. For projectCode → piPid, the determinant is not a superkey and piPid is not part of any candidate key. Strongest normal form: 1NF. projectCode is only part of the composite key but determines the non-key attribute piPid, so the relation violates 2NF.
- BCNF decomposition: Project(projectCode, piPid) and ProjectParticipation(pid, projectCode, role). The first stores each project's PI; the second stores each person's role on a project.

- Lossless: Yes. projectCode is a key in Project, so joining the two relations on it reconstructs the original information without spurious rows.
- Dependency preservation: Yes. projectCode→ piPid holds in Project, and (pid, projectCode) → role holds in ProjectParticipation.

C)

- The only candidate key is $(modelId, certCode)$.
- The relation is not in BCNF because $certCode \rightarrow \{certTitle, issuingAuthority\}$ holds, but $certCode$ is not a superkey. It is not in 3NF because neither $certTitle$ nor $issuingAuthority$ is part of a candidate key. The FD is a partial dependency, so the relation violates 2NF and its strongest normal form is 1NF.
- Decomposition: $R_1 = (modelId, certCode)$ and $R_2 = (certCode, certTitle, issuingAuthority)$.
- The decomposition is lossless because the intersection, $certCode$, is a key of R_2.
- The decomposition preserves the FDs: $certCode \rightarrow \{certTitle, issuingAuthority\}$ holds in R_2, and $(modelId, certCode)$ is the key of R_1.

D)

- The only candidate key is (supp Id, cons Id, lab Id)
- The relation is NOT BCNF
- The FD supp Id → supplier Name violates BCNF. Supp Id is not a superkey because it only determines itself and the supplier name. It is also not 3NF because SupplierName is not part of the candidate key. Since Model Id is atomic, we can conclude that the strongest form is 1NF
- Decomposition: R1= (supp Id, cons Id, lab Id, unit Price) R2=(supp Id, supplier Name)
- Decomposition is lossless because supp Id (the intersection between R1 and R2) is a key of R2
- FDs are preserved: (supp Id, cons Id, lab Id) → unit Price) holds in R1 and supp Id → supplier Name holds in R2

E)

- The only candidate key is the key resId.
- projectCode→projectTitle is not trivial the projectCode is not a superkey so the relation is not BCNF.
- the relation responsible for the violation is projectCode→projectTitle, is because it is not trivial, and projectCode+ =projectCode,projectTitle it didn't identify the relation so it is not in BCNF.we have projectCode→projectTitle and the project-Code is not a superkey and the only candidate key we have is resId, which mean the projectTitle is not a prime attribute and therefore the relation violates the 3NF,that mean that the strongest normal form is 1NF (we didn't see the 2NF,and all attributes are atomic).

- WE put the first relation R1 = projectCode,projectTitle and R2 = resId, projectCode, madeByPid, status,the first relation has a projectCOde is the candidate code cuz is minimal (one attribut) and determine the relation,so R1 satisfies BCNF.the second relation has resId+ = R2 which mean resId is a candidate key(one attribute determine the relation) R2 satisfies the BCNF.
- the intersection between R1 and R2 is projectCode and projectCode→ R1 so the intersection is R1 so the join is lossless.
- .R1 preserve that projectCode→projectTitle and R2 preserve that resId→projectCode,madeByPid which mean the dependency is preserving.

4 Challenges

No significant challenges were encountered during the project work

5 Justification of conversion choices

None.

6 Assumptions

No assumptions were made.